Earth and Planetary Science Letters, 103 (1991) 10–26
Elsevier Science Publishers B.V., Amsterdam

[XLEP]

Paleopositions of the Saxothuringian (Northern Vosges, Pfalz, Odenwald, Spessart) in Variscan times: paleomagnetic investigation

Jean-Bernard Edel [a] and Frank Wickert [b]

[a] *Institut de Physique du Globe, 5 rue Descartes, F-67084 Strasbourg Cedex, France*
[b] *Geologisches Institut, Universität Karlsruhe, Kaiserstr. 12, D-7500 Karlsruhe, FRG*

Received September 14, 1990; revised version accepted December 12, 1990

ABSTRACT

In the Saxothuringian zone, thermal demagnetization of 581 specimens of Late Devonian–Early Carboniferous gabbros, diorites, granodiorites, gneisses and schists yields different, time dependent paleomagnetic directions. The oldest C_n directions (VGP – northern Vosges: 19° N, 222° E; Spessart: 27.5° N, 227° E) were emplaced during crystallization of the Visean plutonics, in the time range 340–330 Ma. C_p directions (N.V: 47° N, 241° E), mostly characterized by low unblocking temperatures, are interpreted as slightly younger overprints or as resultant vectors of C_n and present day directions. The B directions (northern Vosges: 14° N, 132° E; Odenwald: 6° N, 128° E) exist as primary magnetizations in Middle–Late Carboniferous rhyolites and as overprint in the Late Devonian–Early Carboniferous plutonics. The A′ directions (Sp: 28° N, 175° E; Od: 26° N, 168° E) correspond to overprints emplaced later in the Late Carboniferous. The A directions (northern Vosges: 49° N, 155° E; Pfalz: 45° N, 154° E; Odenwald: 48° N, 151° E; Spessart: 47° N, 165° E) are consistent with the Permian–Early Triassic European directions and are present as primary directions in Permian volcanics. The Late Carboniferous–Permo-Triassic overprints are mainly characterized by high unblocking temperatures due to the presence of secondary hematite. The deviations from C to B and from B to A′ are interpreted as due to two successive block rotations, the first 90° clockwise or counterclockwise, the second 40° clockwise, and the deviation from A′ to A to a 2500 km northward drift of the Saxothuringian.

1. Introduction

The Variscan belt, with its bivergent architecture, results from the tectonic interactions between Gondwana and Laurasia from Devonian to Early Permian. In the Mid-European Variscan a central crystalline ridge with Proterozoic and early Paleozoic metamorphics (Moldanubian-Mo, Fig. 1) is bordered on the north by an Early Carboniferous magmatic arc of active continental margin [1,2]. On the north of this so-called Saxothuringian zone (Sa; [3]), the Phyllite Zone (Ph), with Devonian series [4], forms the border between the Saxothuringian and the Rhenohercynian (Rh) which is characterized by Devonian to Early Carboniferous sedimentary rocks and volcanics. Late Carboniferous to Permian red beds and volcanics overly unconformably the deformed pre-Late Carboniferous sequences.

The contacts between the three zones seem to be of tectonic origin. The Mo–Sa border, the so-called Lalaye–Lubine–Baden-Baden (LLBB) fault was a Late Devonian–Early Carboniferous SE-dipping thrust fault, overprinted by a Carboniferous right-lateral oblique strike-slip fault [5]. The Ph zone acted as border between Sa and Rh. In the Ph zone, Early Carboniferous NW-vergent thrusting was followed by right lateral strike-slip faulting and Permian–Tertiary normal faulting. Within the Sa, Early Carboniferous convergence and Late Carboniferous–Early Permian extension caused predominantly NE–SW-trending structures (mylonitic foliations, folds, cataclastic strike-slip faults and normal faults). Intrusions of mafics followed the regional structures.

The paleomagnetic results obtained in different massifs of the Variscan belt as Armorican Massif, Central Massif, Vosges, Schwarzwald and Sude-

0012-821X/91/$03.50 © 1991 – Elsevier Science Publishers B.V.

tians, demonstrate that (1) large-scale magnetic overprints have affected the Hercynides and their environment in the Late Carboniferous–Permian [6–10] and (2) large bloc rotations occurred during the Carboniferous. The youngest, in Westphalian–Stephanian times has affected all investigated massifs [7–10]. During the late Visean–Namurian, the Moldanubian southern Vosges and Central Massif and the Saxothuringian northern Vosges have undergone a first rotation phase [8,9,11,12]. The purpose of this study is to establish whether other massifs of the Saxothuringian were involved in the corresponding tectonics and to propose a model of evolution of the zone, based on paleodirections and paleolatitudes.

2. Geology

The northern Vosges are composed of Early Paleozoic monotonous clastics and Middle Devonian to Early Carboniferous sediments and volcanics [13,14]. These sequences are intruded by Early Carboniferous unfoliated I-type granites and granodiorites emplaced around 340 Ma [11] and by Late Carboniferous S-type granites. Stephanian and Permian continental sediments and felsic volcanics overly unconformably the Devonian–Early Carboniferous rocks. Early Carboniferous NW–SE-directed convergence led to the development of E–W- to ENE–WSW-trending folds with a SE-dipping axial plane cleavage [5]. Right lateral strike-slip faulting along E–W- to ENE–WSW-trending wrench-faults outlasted thrusting, especially along the Lalaye–Lubine zone. Deposition of Late Carboniferous–Permian sedimentary rocks and volcanics over folded sequences is concentrated along these strike-slip faults.

In this study, previous results [11] obtained on Early Carboniferous microdiorites, diorites and granodiorites from the Champ du Feu massif and on Late Carboniferous rhyolites related to the Kagenfels granite, are taken into account. Additional sites are collected in microdioritic dikes and inclusions (VS 44–45, 50, 51, 53) and in a granite (VS 41), (Fig. 1). One additional site (VN 24b) was sampled in Ordovician–Silurian weakly schistous sediments, close to the southern border of the plutons. North of the massif, Frasnian and Visean black (VN 56), red (VN 46) and green schists (VN 47, 58, 60, 61) were collected in the hope that they have not undergone complete remagnetization in the Permo-Trias. VN 47b represents a dike intruded into VN 47.

The northern Schwarzwald consists of several crustal blocks, characterized by different stratigraphic, tectonic and metamorphic evolutions (LP/HT gneisses, MP/HT micaschists, LP/LT mylonitic metasediments and metavolcanics, and sheared granites of S- and I-type character). The folded sequences are uncomformably overlain by Late Carboniferous–Permian red beds. Kinematic indicators suggest right-lateral shearing of metasediments and S-type granite and left-lateral shearing of I-type granite during Early Carboniferous [15].

In the Pfalz, small outcrops under Triassic cover demonstrate the presence of gneisses (*d, Fig. 1), schists and sandstones, probably of Early Carboniferous age (*b,e), unfoliated granodiorites and granites (*b,c,d) including lamprophyric dikes (*d) and Permian red beds and volcanics. Foliation in gneisses trends E–W, while bedding in schists and sandstones trends NE–SW. Sampling was performed in granodiorites (Pf 2), in gneisses and their lamprophyric intrusions (Pf 3), and in Permian melaphyres (Pf 1, 4). Results from the Windstein granodiorite (Ws 1, 2; 14) dated respectively at 339 ± 11 Ma on amphibolites and 342 ± 10 Ma on biotites [16] are also taken into account.

The Odenwald is characterized by the presence of gneisses, metasediments and amphibolites intruded by Late Devonian gabbros and Early Carboniferous diorites to granites [17,18]. The radiometric ages around 342 Ma (K-Ar) obtained on hornblendes of various plutonics are close to the ages measured on similar rocks from the northern Vosges and Windstein. Intruded dioritic bodies trend NE–SW, while intrusions of granites are more disordered (Fig. 1). Diorites and granites are in turn cut by lamprophyric dykes. The occurrence of Permian red beds and felsic volcanics is restricted to the Rhinegraben border. Regional trend of foliation is NE–SW, steeply dipping. Paleomagnetic sampling concerns metamorphics (Od 3, 4, 7, 10, 13), amphibolites (Od 1, 4, 16b), gabbros (Od 12, 14, 15, 24, 25), diorites (Od 8, 9, 19, 22), granodiorites (Od 2, 5, 6, 18), granites (Od 16, 17, 26) and lamprophyric dikes (Od 20,21) respectively intrusive into Od 19 and Od 6.

In the Spessart, medium-grade micaschists,

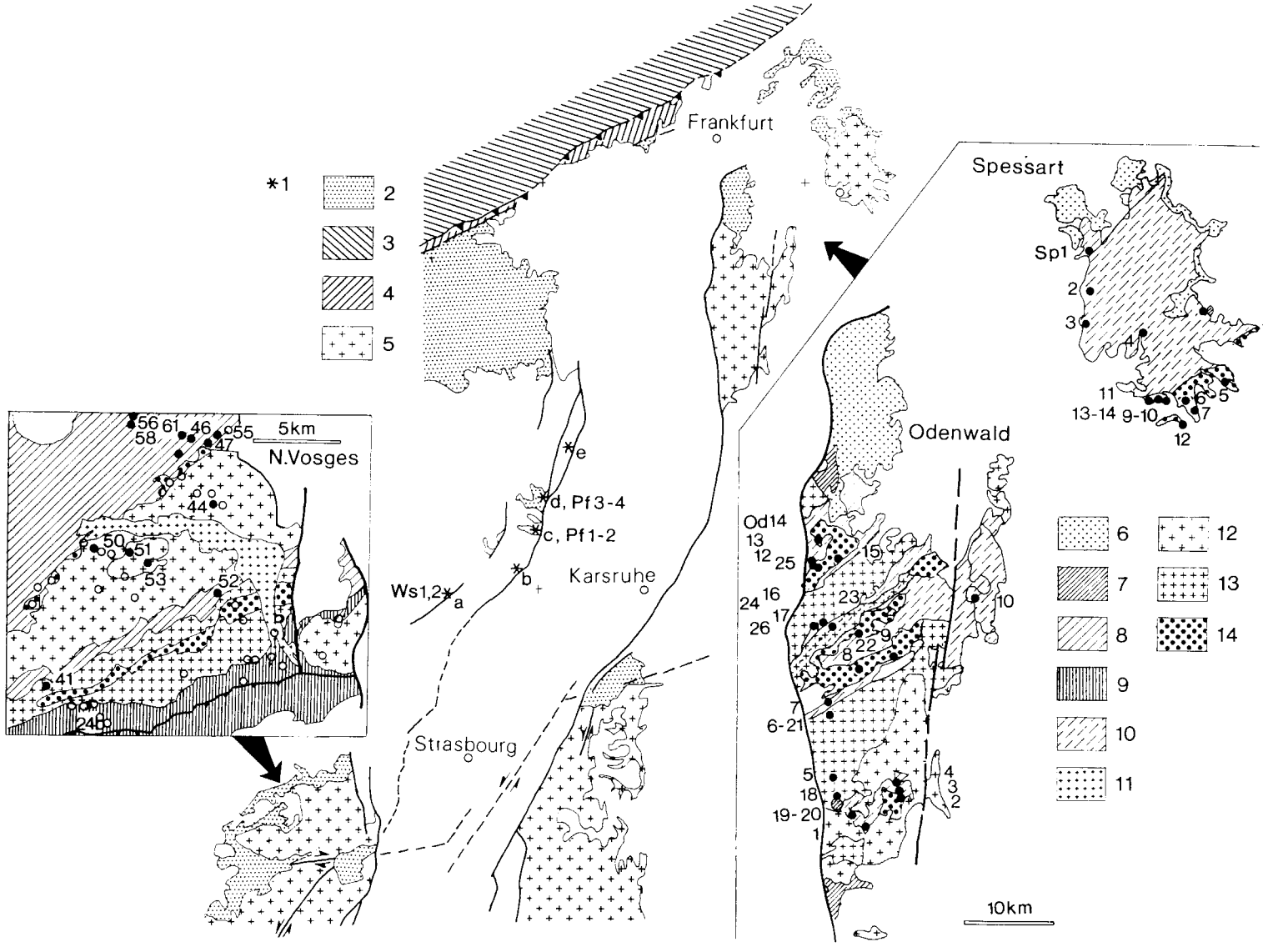


Fig. 1. The Saxothuringian outcrops in the upper Rhinegraben area. Location of paleomagnetic sites. *1* = small basement outcrops of Pfalz and Windstein; *2* = Permian; *3* = Rhenohercynian; *4* = Phyllite zone; *5* = Moldanubian; *6*, *7* = Permian sediments, volcanics; *8* = Devonian-Dinantian sediments and volcanics; *9* = Cambrian to Silurian schists; *10* = metamorphics; *11* = Late Carboniferous granites; *12* = granites; *13* = granodiorites; *14* = diorites and gabbros.

paragneisses, orthogneisses and amphibolites are intruded in the southern part by diorites and granodiorites. Lamprophyre dikes intrude basement and mafic stocks. Permian red beds and volcanics lie unconformably on the folded sequences. Rb-Sr ages of orthogneisses are 391 ± 11 Ma and 402 ± 8 Ma [19,20]. Samples were collected in gneisses and amphibolites (Sp 2, 3, 4, 9) of northern Spessart and in diorites (Sp 5, 6, 10, 11, 14), in lamprophyres (Sp 9, 13) respectively intrusive into diorites Sp 10 and Sp 14, and in an aplite granite (Sp 12) from southern Spessart.

3. Magnetic properties

A paleomagnetic site represents an exposure of at least 10 m^2, for instance a quarry. The oriented samples were drilled or collected as handsamples and the 25 mm diameter cores are cut into two to four, 22 mm long specimens. Modified Digico equipment was used for determination of the susceptibility, anisotropy of susceptibility and remanent magnetization.

With exception of the Paris Basin anomaly, the Saxothuringian demonstrates the most intense distribution of magnetic anomalies within the European continental plate (aeromagnetic maps published by the Bundesanstalt für Bodenforschung, FRG and by the Bureau de Recherche Géologiques et Minières, France). The distribution of the *magnetic susceptibility* and *remanent magnetization* in rock specimens reveals the latest Devonian–Early Carboniferous mafic plutonics as the principal responsible for the anomalies. On outcrops, the magnetic bodies can be identified as gabbros (Frankenstein gabbro, Odenwald), diorites (Odenwald and northern Vosges), and as granodiorites (Odenwald, Windstein and northern Vosges). Late Carboniferous dykes from the Spessart are also characterized by high values of induced and remanent magnetizations. Susceptibilities and remanent magnetizations respectively ex-

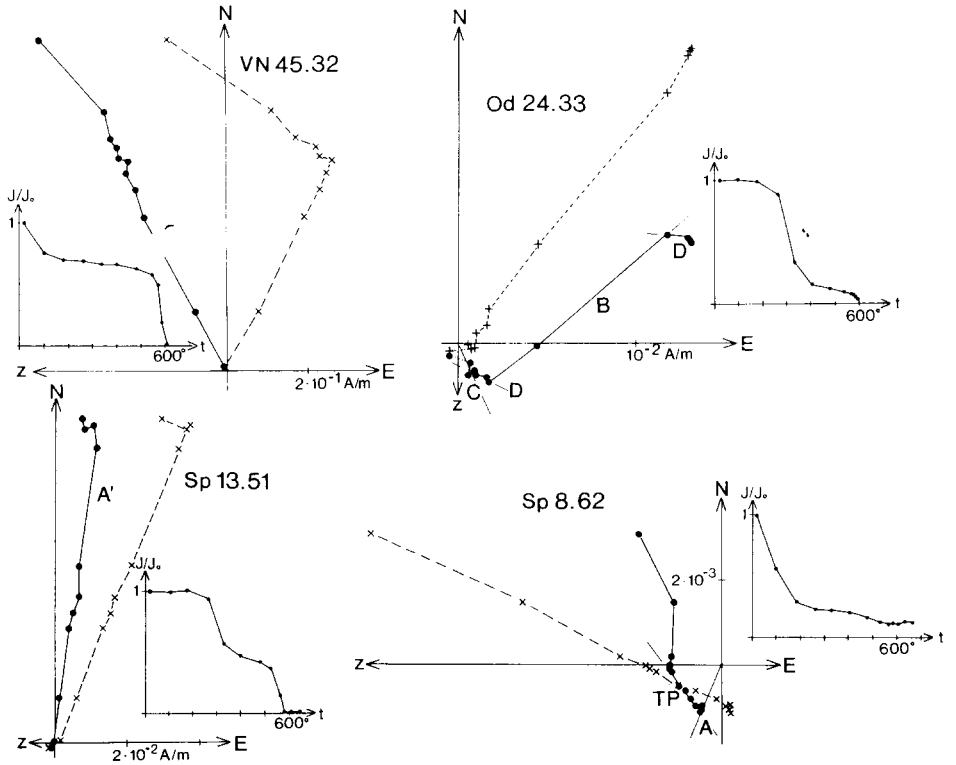


Fig. 2. Typical thermal demagnetizing curves and orthogonal projections in: a microdiorite from northern Vosges (VN 45.32); a gabbro from Odenwald (Od 24.33); a lamprophyre dyke (Sp 13.51); and a Late Carboniferous–Early Permian ignimbrite (Sp 8.62) from Spessart. Dots are in the horizontal plane, crosses in the vertical plane. The directions are labelled TP, A, A', B, C, D.

ceed 10^{-4} SI and 10^{-1} A m^{-1} in gabbros and diorites from Odenwald and northern Vosges. In the northern Vosges, the Late Devonian–Early Carboniferous mafic volcanics display high values of magnetization comparable to those of the mafic plutonics [11].

The Koenigsberger ratio Q (ratio of remanent/induced magnetization) stays in the range 0.1–1, with a maximum around 0.4 for rocks containing magnetite. When hematite is the dominant magnetic mineral, as for instance in rhyolites (northern Vosges) or in metamorphic schists (Spessart and northern Vosges), the susceptibilities remain weak (10^{-4}–10^{-3}), while the remanent magnetizations can reach 1.0 A m^{-1}. In such rocks, the Q ratio mostly exceeds 1.

By extrapolation, most of the magnetic bodies of the Saxothuringian evidenced on the magnetic maps can be interpreted as Late Devonian–Early Carboniferous mafic intrusives and/or as associated extrusives. They show a predominant Variscan N50–60° trend in northern Vosges, Spessart and Odenwald that excludes significant relative rotations of the different massifs.

4. Paleomagnetism

Paleozoic rocks are mostly characterized by multicomponent remanent magnetizations. Usu-

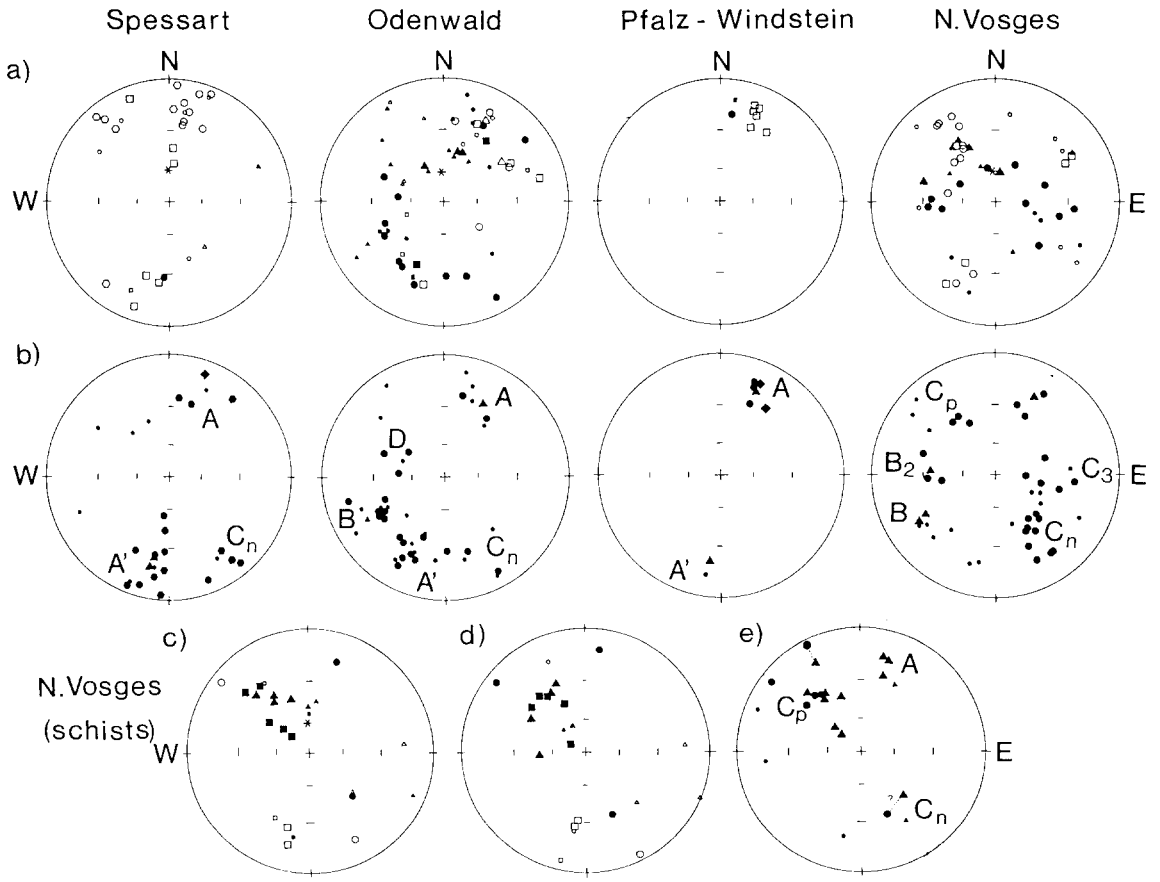


Fig. 3. Stereographic projection of the mean site directions obtained after thermal demagnetizing in plutonics, metamorphics and volcanics of Spessart, Odenwald, Pfalz and northern Vosges. (a) All *in situ* low-temperature (triangle), intermediate-temperature (dots) and high-temperature (square) components are plotted. (b) After removing the post-Early Triassic overprints, the directions are plotted in the lower hemisphere. Losanges = Early Permian volcanics, triangles = Late Carboniferous dykes, dots = metamorphics and Early Carboniferous plutonics, hexagons = results from Bosum et al. [25]. (c) *In situ*, low-temperature (triangles), intermediate- (dots) and high-temperature (squares) directions obtained in Cambrian to Visean schists of the northern Vosges. (d) Directions in schists after tilt correction. (e) The pre- and post-tilting directions are plotted in the lower hemisphere.

ally two or three different components were isolated by stepwise demagnetization of 581 specimens. As it has been observed in other Paleozoic areas, thermal demagnetization is much more efficient than the alternating field procedure.

Some sites as the gabbro Od 24a (Od 24.33, Fig. 2), are characterized by a polyphased magnetic history. In plutonics, magnetite is the dominant carrier of remanent magnetizations, as show the maximum unblocking temperatures around 580 °C (VN 45.32, Od 24.33, Sp 13.51, Fig. 2). When magnetite is present, low unblocking temperatures in the range 300–400 °C of unclear origin can often be observed (sulfides or maghemite?; Od 24.33, Sp 13.51, Fig. 2). Hematite carries the characteristic remanent magnetization in felsic dykes (rhyolites from the northern Vosges), in metamorphic schists (northern Vosges and Spessart), in Permian volcanics ($t_b > 660$ °C in Sp 8.62, Fig. 2) and in numerous plutonics considered as remagnetized.

On stereograms, the mean directions of the

TABLE 1

Spessart (50 ° N, 9.2 ° E), paleomagnetic results [a]

Site	Formation	t_b	N/N_0	n	D_0	I_0	a	K	α_{95}	VGP	
										°N	°E
Sp 8	ign.Tuf., Permian	> 660	6/6	8	199	−7	A	85	6		
Sp 2	amphibolite,	580	4/9	5	184	27	A′	58	10		
	quartzite	300–400	2/9	2	69.5	−14	B	558			
		350–580	2/9	2	306	21	C	29			
Sp 3	amphibolite	300–400	3/3	2	141	−41	C	216.5			
		> 660	2/3	4	339	−7	C	108	9		
Sp 4	metagranite	> 660	6/6	7	5	−43	A′	204	4		
Sp 5	diorite	> 660	3/5	6	8	−55.5	A′	55	9		
		580–600	2/5	4	160	−39	C	44	14		
Sp 6	diorite	580	5/6	8	11	−25	A′	111	5		
Sp 7	diorite	580	2/5	3	10	−17	A′	36	21		
		580	1/5	3	330	−15	C	133	11		
Sp 9	metagranite 1	580	3/9	5	22	−4	A′	40	12		
	lamprophyre 2	580	5/9	6	11	−24	A′	140	6		
	close to Sp 10	> 620	2/9	3	203	−15	A	61	16		
Sp 10	diorite	580	4/4	6	16	−6	A′	270	4		
Sp 11	diorite 1	580	2/6	3	21	−3.5	A′	294	7		
	2	580	4/6	6	25.5	−24	A′	88	7		
Sp 12	aplite granite	> 660	3/3	3	196.5	−26	A	82	14		
Sp 13	lamprophyre intrusive into Sp 14	580	5/5	7	13	−17	A′	540	2.5		
Sp 14	diorite	580	4/6	5	9	−12	A′	183	6		
		> 660	4/6	6	187	−24	A	248	4		
Means											
C				2	334.5	−11		92.6	26	30	219
C (including C Bosum et al. [25])				5	327	−11		78	9	27.5	227
A′: lamprophyre dykes				2	12	−21		250		28	176
A′: including overprints				8	14	−17.5		52		30	173
A′: including overprints				12	13	−21.5		24	9	−28	175
A′: Sp 4, 5				2	6	−49		82	28	10	184
A				4	196.5	−18		56	12	−47	345

t_b = maximum unblocking temperature; N = number of samples which carry the considered direction; N_0 = total number of samples; n = number of specimen which carry the considered direction; D_0 and I_0 = *in situ* declination and inclination; D_c and I_c = tilt-corrected declination and inclination; a = direction label; K, α_{95} = Fisher statistic parameters.

TABLE 2

Odenwald (49.7° N, 8.75° E), paleomagnetic results [a]

Site	Formation		t_b	N/N_0	n	D_0	I_0	a	K	α_{95}
Od 1	amphibolite		350	2/2	4	333	59	TP	305	5
Od 2	granodiorite		580	5/7	8	63	−28	B	36	9
			350, 580	2/7	3	31	−14	A′	13	35
			350–400	3/7	3	16	45	TP	160	10
Od 4	amphibolite		350	2/7	3	332	−6	C	119.5	11
	metagranite		350, 580	2/7	3	20.5	22	A	27	24
Od 5	granodiorite		> 650	2/2	2	202	23.5	A′		
			400	2/2	5	13	49	TP	92	8
Od 6	granodiorite		580, > 650	3/7	5	76	−13	B	44	11
			580, > 650	6/7	10	24	−21.5	A′	29	9
Od 8	diorite		400, 580, > 660	4/5	10	60	−27	B	27	9
			400, > 660	2/5	2	216.5	−34	A	201	18
Od 9	diorite		400–580	2/7	4	244	31.5	B	27	18
			580	3/7	4	28	−12	A′	50	13
		N + R	580, > 660	4/7	7	35.5	29	A	37	10
Od 12	gabbro		400, 580	3/7	8	162.5	26	C	79	6
			500, 580	3/7	4	239	33	B	36	15
			550–580	5/7	9	199	36	A′	34	9
			400	1/7	3	305	35	C-D	23	26
Od 13	hornfelses		350	2/2	3	7	−22	A′	21	27
Od 14	gabbro		350	2/3	4	329	8	C	26	18
		N + R	350, > 600	2/3	5	250	−56	?	141	6
Od 15	gabbro		580	3/6	8	293	36	C-D	20	12.5
			600	2/6	3	241	37	B	16	31
			580	2/6	3	20	−38	A′	106	12
Od 16	granite	N + R	580, > 620	6/9	8	203.5	22	A′	28	10
			580	2/9	4	69	−19	B	19	21
	amphibolite + granite	N + R	330, > 580	2/9	6	312	−46	C	54	9
Od 18	granodiorite		> 660	3/7	4	193	−22	A	31	16.5
Od 19	diorite		300–400	5/6	6	28	−17	A′	60	9
Od 20	dyke in Od 19		300–400	3/3	5	333	54	TP	38	12
Od 21	dyke in Od 6		540	3/4	5	21	43	TP	43	12
			580	3/4	6	28	21	A	103.5	6.5
			580	1/4	3	241.5	20	B	18	29
Od 22	diorite		580	5/6	7	212	27	A′	37	10
			350	1/6	2	315	20.5	C	85	27
Od 23	granite		580	7/7	11	178	29	A′	38	7
			550	2/7	4	137	30	C	51	13
			400	1/7	2	237	10.5	B	96	25
Od 24a	gabbro		550	4/6	5	126	−51	D	50	11
			580	5/6	9	151	8	C	46	7.5
			350	3/6	7	55	−29	B	25	12
24b	gabbro		280, 580	2/3	4	27	−28	A′	68	11
Od 25	gabbro		> 580	4/5	8	253	32.5	B	34.5	9
			550	4/5	8	276	50	D	66	7
			580	2/5	3	223	45		627	5
Od 26	granite		580	3/3	4	9	−21.5	A′	60	10
			330	2/3	3	348	−21	C-A′	1772	3

TABLE 2 (continued)

Site	n	D_0	I_0	K	α_{95}	VGP	
						°N	°E
Means							
D	3	292	46	39	20	34	282
	5	295	45	47	11	36	279
B	6	64.5	−27	60	9	5	127
	11	63.5	−26	59	6	6	128
A′	6	26	−23	72	8	25	167
	14	19	−23	34.5	7	26	168
A	2	200.5	−22	67	31	−48	338
	6	205	−24	54	9	−48	331

[a] For explanations, see Table 1.
The second mean takes into account directions from the one or two samples per site.

different components evidenced by thermal demagnetization, show a scattered distribution, particularly in the Odenwald and the northern Vosges (Fig. 3a; Tables 1–5). Because of the occurrence of normal and reversed directions, the scatter is reduced after plotting all directions in the same hemisphere (Fig. 3b). Here, the lower hemisphere was arbitrarily chosen. Different clusters arise, labelled TP, A, A′, B, B′, C, C_3 directions, according to the nomenclature adopted in previous studies [8,9,11].

In the northern Vosges and the Odenwald, a few low-temperature (300°–500°C) components with directions more or less close to the present field direction (TP) were isolated. They are interpreted as overprints, emplaced subsequently to the Early Trias.

5. The A directions

The A directions are characterized by SSW declinations and shallow inclinations which are very common in the Variscan belt and surrounding areas. They were found in all investigated outcrops of this study (Fig. 3). The A components are interpreted as primary magnetizations in Permian ignimbrites from the Spessart (Sp 8). In the Late Stephanian–Early Permian basalts from the Pfalz (Pf 1, 4) the hematite blocking temperatures indicate a remagnetization.

TABLE 3

Pfalz (49.2°N, 8°E) and Jaegertal (49.5°N, 7.7°E), paleomagnetic results [a]

Site	Formation	t_b	N/N_0	n	D_0	I_0	a	K	α_{95}	VGP	
										°N	°E
Pf 1	melaphyre, Permian	580, 680	4/4	7	203	−12	A	57	8		
Pf 4	melaphyre, Permian	680	3/3	4	214	−23	A	272	5.5		
Pf 2	granodiorite	> 650	2/6	3	188	12	A′	98.5	12		
		> 650	3/6	3	200	−12	A	165	10		
Pf 3	dykes	580	3/10	5	187	20	A′	20	8.5		
		> 620	6/10	8	203	−17	A	84	6		
Ws 1	granodiorite	> 650	7/7	14	200	−15	A	68	5		
Ws 2	granodiorite	670	5/5	9	202	−25	A	61	6		
Means											
A′				2	187.5	16		202		−32	359
A				6	203.5	−17		119	6	−45	334

[a] For explanations, see Table 1.

TABLE 4

Northern Vosges — Champ du Feu Massif (48.4° N, 7.25° E), paleomagnetic results [a]

Site	Formation	t_b	N/N_0	n	D_0	I_0	a	K	α_{95}	VGP	
										°N	°E
VN 41	granite	> 660	4/7	5	197	−28	A	139	6		
VN 44, 45	microdiorite	300–400	6/13	7	328	43	C_p	28	11		
		300–400	2/13	3	196	17	A′	1050	4		
		580	13/13	15	335	−22	C_n	63	5		
VN 50	microdiorite	300–400	5/6	6	322	32	C_p	103	7		
	(dyke)	580	6/6	10	314	−41	C_n	125	4		
		> 620	4/6	8	206	−33	A	270	3		
VN 51	microdiorite	300–400	3/3	7	329	31	C_p	25	12		
	(inclusion)	580	3/3	5	325	−32	C_n	119	7		
VN 52	volcanics,	580	2/6	3	218	31	A′	38	20		
	Devonian	> 650	3/6	4	210.5	−15	A	164	7		
VN 53	microdiorite (inclusion)	580	5/5	5	329	−37	C_n	1337	2		
Means, including results from Edel et al. [11]											
C_p				3	326	32		54	17	48,	241
C_n				10	326	−29		40	8	19,	222
C_3				6	92	41		24	14	−16,	260
B_1				5	54	−21		153	6	14,	132
B_2				4	272	36		60	12	−16,	289
A′				2	206	24		22	55	−24,	339
A				5	203	−24.5		78	9	−49,	331

[a] For explanations, see Table 1.

Close below the volcanics of the Pfalz, granodiorites (Pf 2), metamorphics and lamprophyres (Pf 3), mostly display the same A directions. Granodiorites (Pf 2), also those from Windstein (Ws 1,2), show clear evidence of hydrothermal alteration. The directions of remanent magnetization [21] are consistent with those of neighbouring Permian volcanics (Pf 1, 4, Sp 8, the Nideck ignimbrite in the northern Vosges, [22]) but biotite and amphibole from Ws 2 display concordant K-Ar crystallization ages of respectively 339 ± 11 and 342 ± 10 Ma [16]. The younger age of the remanent magnetization is believed to be due to secondary hematite visible on thin sections in small cracks and along mineral boundaries. The same conclusion can be drawn in the northern Vosges, Odenwald and Spessart, where the A components are mostly characterized by high unblocking temperatures (> 600 °C). The Late Devonian–Early Carboniferous red and grey schists of the Bruche valley (northern Vosges) are also characterized by A directions with high unblocking temperatures.

Most of the A components are reversed, indicating that they were emplaced during the reversed Kiaman field interval. Some normal components measured in the Odenwald and northern Vosges suggest that overprinting lasted up to the Triassic.

6. The A′ directions

The declinations do not differ significantly from the A components but for the SSW directions the inclinations become positive (and conversely; Fig. 3). A′ normal directions are present in N–S-trending lamprophyre dykes of the Spessart (Sp 9, 13) and of the Pfalz (Pf 3). In the latter site, only the freshest specimens show the A′ component, the other display the A direction found in the overlying Permian volcanics. The Spessart lamprophyres contain a single magnetic direction A′, characterized by both low (330 °C) and intermediate (580 °C) unblocking temperatures, the latter revealing magnetite as carrier of the remanence (Sp 13.51, Fig. 2). The omnipresence in the metamorphic (Sp 9a) and dioritic (Sp 10, 14) hostrocks of the same A′ directions implies that they have probably been overprinted by the late magmatic

TABLE 5

Northern Vosges — sedimentary host rocks, paleomagnetic results [a]

Site	Formation	t_b	N/N_0	n	D_0	I_0	D_c	I_c	a	K	α_{95}	VGP °N	VGP °E
S. of the Champ du Feu massif													
VN 24b	schist,	670	5/9	7	313	19.5			C_p	29	11		
	Silur.–Ordov.						310	31	C_p	48	8		
		670	3/9	4	312	68			D?	44	14		
							297	75		40	14.5		
N. of the Champ du Feu massif													
VN 46	schist, Frasnian	> 620	5/5	8	197.5	−25			A	35	9		
							186	−31		33	10		
VN 61	schist, Frasnian	470	5/5	9	317	25.5	329	32	C_p	33	9		
		580	3/5	7	308	−6	308	6	C_n?	55	8		
VN 47	schist, Visean	400	2/4	4	8	45	333	63.5	TP	60	12		
		560–580	4/4	8	136	38.5	157	32	C_n	95	6		
VN 47b	dyke, intr.	300, 580	4/5	7	325	37	302	36	C_p	84	7		
	in VN 47	580	3/5	3	358	50	315	62	TP	62	16		
		580	1/5	2	328	−23	338	−14	C_n	26	51		
VN 55	schist, 1	550–580	4/10	10	153	−13	153	−3	C_p	51	7		
	Visean 2	580	2/10	3	113	−6	113	0	C_p	60	16		
		> 650	7/10	17	194	−14	193	−5	A	87.5	4		
VN 56	schist,	350, 550	3/4	5	17	16	8	11	A	39	12		
	Visean	> 600	2/4	4	208	−28	188	−27	A	33	16		
VN 58	schist,	350, 580	3/4	9	328	34	267	50	C_p	24	10		
	Visean	550, 580	4/4	6	359	56	214	62	TP	94	8		
VN 60	schist,	280	2/4	7	15	67			TP	97	6		
	Visean	450–500	4/4	13	341	41			C_p	59	5		
							337	29.5		14	11		
Means													
C_p				5	328.5	30				36	13	49,	237
							318	32.5		8	27		
C_n (47)				1	136	38.5	157	32					
C_p [b]				8	325	31				48	8	47,	241
A				4	199	−21				84	10	−49,	338
							189	−19		40	14.5		

[a] For explanations see Table 1.

[b] All sediments, north and south of Champ du Feu, including previous results [11].

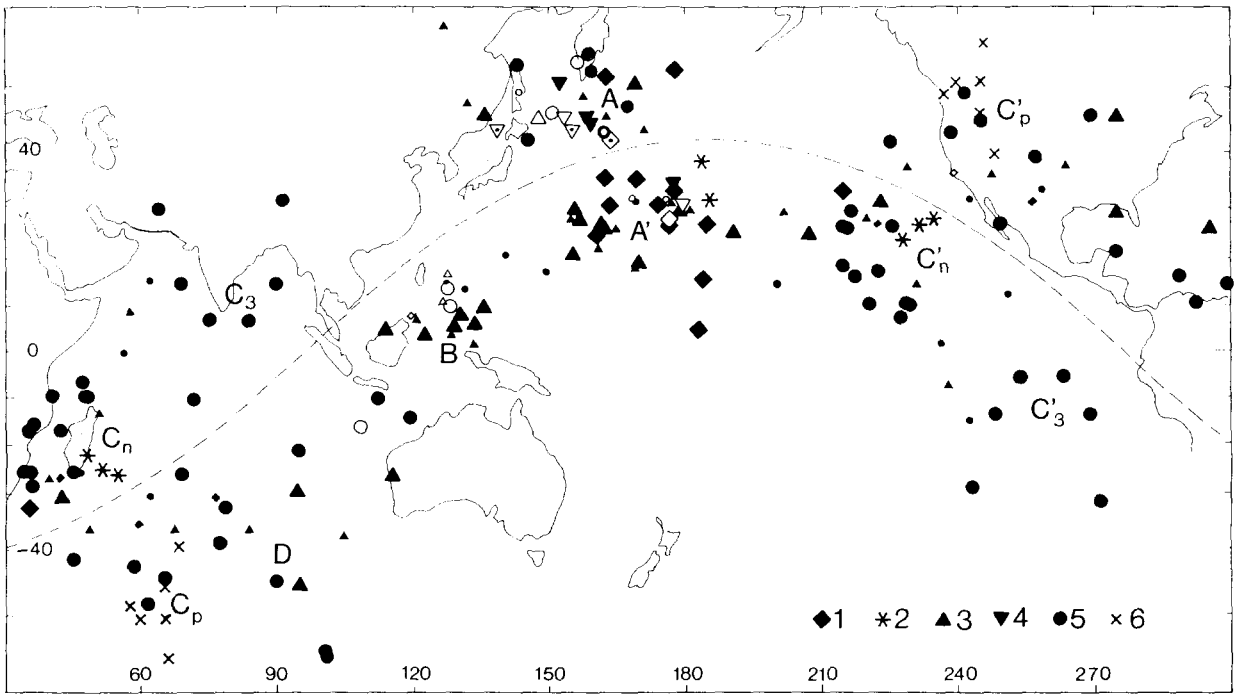
intrusions. Even diorites distant from the dykes display this direction (Sp 6, 7).

Two sites, a diorite quarry (Sp 5) and a metagranite quarry (Sp 4), show A′ directions with steeper inclinations around −50°. They are characterized by hematite unblocking temperatures.

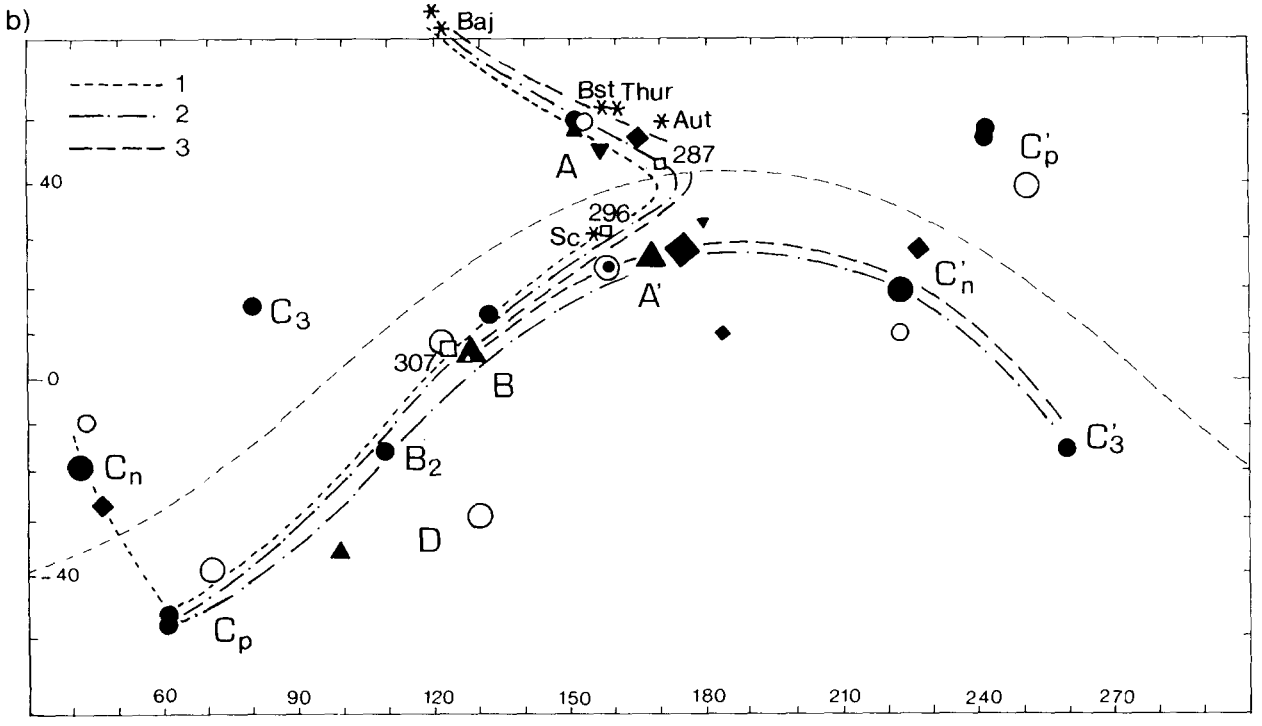
In the Odenwald, the A′ directions are as much normal as reversed, without any geographical or petrological preference. Magnetite and/or hematite are the carriers of the remanence. Hematite being not primary in the mafic plutonics, the A′ magnetizations are considered as overprints. In four sites (Od 5, 6, 9, 22) showing A′ directions, K-Ar dating on hornblende has yielded ages in the range 342–337 Ma, and ages of 330–329 Ma on biotite [20]. The overprints with hematite unblocking temperatures are supposed to result from low-temperature hydrothermal processes. Consequently, the biotite ages, corresponding to the isotherm 300°C are a lower limit for emplacement of the A′ remagnetizations.

As there is no direct age available for the A′ magnetizations in the Saxothuringian, attempt is made to date them by correlations with results from other Variscan massifs. Close directions were obtained in a rhyolite dated at 296 ± 5 Ma from

a)


A
A'
B
C3
Cn
Cp
C'p
C'n
C'3
D
40
0
-40
60
90
120
150
180
210
240
270
◆1 ✱2 ▲3 ▼4 ●5 ×6

b)


Baj
Bst Thur
Aut
287
296
Sc.
307
A
A'
B
B2
C3
Cn
Cp
C'p
C'n
C'3
D
1
2
3
40
0
-40
60
90
120
150
180
210
240
270

Central Schwarzwald [8,23] (Fig. 4b) and in a Late Carboniferous granite cooled at 300 Ma from NE Central Massif [8]. A′ directions were also found in Stephanian-Early Permian layers from the Schopfheim basin, in the southern Schwarzwald [24]. Consequently, a Stephanian age around 300 Ma is proposed for the A′ directions.

7. The B directions

With inclinations comparable to those of the A′ components, the B directions are characterized by deviated WSW declinations (and conversely). They occur in Odenwald and northern Vosges intrusives (Fig. 3). In the southern Champ du Feu massif (northern Vosges), they were identified as primary magnetizations of normal polarity carried by primary hematite, in a Late Carboniferous rhyolite associated to the Kagenfels granite, and as overprint in the dioritic and granodioritic hostrocks [11]. The Early Carboniferous plutonics also display westerly directions, labelled B′, and interpreted as B directions, tilted or rotated prior to the rhyolitic intrusion.

The B components of the Odenwald are as much normal as reversed and fairly antiparallel (Fig. 3). The characteristics of the magnetizations are often similar to those observed for the A′ components. For instance, in the granodiorite Od 6, which contains B components in some samples and A′ in others, the behaviour during thermal demagnetizing is the same; the magnetization shows intermediate to high unblocking temperatures revealing the presence of hematite. The only difference is an apparent greater freshness of the samples containing the B component. With the exception of gabbro Od 24a, which is marked of by its low unblocking temperature (300–350 °C), magnetite and/or hematite carry the B remanences. Hematite being very likely not primary in the Early Carboniferous mafic plutonics, the B components as the A′ can be considered as overprints emplaced after 328–330 Ma which are the ages given by K-Ar dating on biotite.

B directions with normal polarity exist also in the Moldanubian zone; they were previously identified as primary in ignimbrites from central Schwarzwald, dated at 307 ± 5 Ma [8,23] (Fig. 4b), and in Late Visean felsic volcanics from northeastern Central Massif, mostly as post-Late Visean–Namurian folding overprints [8–10]. B reversed components were found as overprints in amphibolites from the northern Central Massif, intruded by granites around 312 Ma [8,9]. In granodioritic hostrocks close to the Kagenfels intrusion (northern Vosges), rejuvenated K-Ar ages of 312, 313, 314, 318, 320 Ma which may be due to the late magmatism, were obtained [11]. Consequently, the B directions likely represent the magnetic field in the time range 320–307 Ma, corresponding to the Namurian-Westphalian.

8. The C directions

Northwesterly directions are frequent in the Champ du Feu massif (northern Vosges) and in the Spessart, where they have been found on not demagnetized NRM [25]. The additional sites confirm the previous results.

In northern Vosges, the C_n directions, characterized by negative inclinations and previously labelled C_1 and C_2 [11], show a nice cluster. They were obtained from subvertical to NW-dipping, microdioritic, and microgranitic belts of the northern Champ du Feu massif and dioritic inclusions in granitic belts. The carrier of the remanence is magnetite, as indicated by the 580 °C maximum unblocking temperatures (VN 45.32, Fig. 2) and

Fig. 4. (a) Mercator projection of the Virtual Geomagnetic Poles of the Spessart (*1*, *2* after [25]), the Odenwald (*3*), Pfalz and Windstein (*4*), northern Vosges (*5*). Crosses (*6*) are low-temperature C_p poles of northern. Vosges. Open symbols represent Late Carboniferous dykes and dotted open symbols Early Permian volcanics. Small symbols correspond to directions measured in specimens from 1 or 2 samples. Because of the unclear polarity of the C and D directions, both poles are plotted. The dotted pole line corresponds to the directions with $I = 0°$ i.e. to the paleoequator. (b) Mean poles obtained in the investigated Saxothuringian massifs (same symbols as above) and in the neighbouring Moldanubian, and the alternative Polar Wander Paths *1*, *2* and *3*. Small symbols represent 1 or 2 sites, intermediate symbols 3 to 9 sites, big symbols, 10 sites and more. Open dots are mean poles from the Moldanubian southern Vosges (in work), open squares are results from dated rhyolites from the Schwarzwald [11,23], stars are reference poles for the Autunian [32], the Thuringian [33], the Late Buntsandstein [34], the Late Bajocian [35,36]. Sc is a mean result for the Stephanian–Rotliegendes of the Schopfheim basin (southern Schwarzwald) [24].

by IRM acquisition curves. A southeasterly C_n direction was also obtained in Visean green schists (VN 47).

C_p NW directions with positive inclinations are also present, as low-temperature magnetization in microdiorites (VN 44–45, 50, 51) and in Devonian–Dinantian green schists north of the plutonic massif (VN 58, 60, 61) and as high blocking temperature magnetization (hematite) in contact metamorphic schists very close to the southern border of the massif (VN 24a, 24b, 25).

The easterly C_3 directions may belong to the C family. They are limited to the northern boundary of the plutonic Champ du Feu massif. In the previous study they were interpreted as rotated or tilted C directions.

With exception of a single dioritic sample, the freshest of Sp 7, the rocks from Spessart, characterized by C_n components, consist of metamorphics: gneisses and amphibolites from the northern part. The rather good clusters obtained on undemagnetized NRM by Bosum et al. [25], can be exemplified by the thermal treatment of Sp 3: from room temperature up to 670 °C, the direction of the remanence remains stable, indicating that hematite is the carrier of the magnetization.

The C directions measured in one sample of the SP 7 diorite is interpreted as having escaped the A′ overprint phase related to the lamprophyre intrusions. Away from the intrusions, and due to hematite being the carrier of the C components, the metamorphics of the northern massif were less affected by the overprints than the diorites.

A few NNW and SSE directions appear also in different rocks from Odenwald (Od 4, 12, 14, 23, 24), often only on a single sample (Od 4, 14, 23). In the gabbros Od 24a and Od 12, characterized respectively by three (B, C, D) and two (A′, C) components, the C component has the highest unblocking temperatures (580 °); it is probably the oldest.

Concerning the age of the magnetizations, the microdioritic sites VN 3 and 4 characterized by C_p directions in the temperature range 20–350 °C and C_p directions up to 580 °, were dated at 328 and 327 ± 10 Ma by K-Ar techniques on amphibole and biotite [11]. The granite with the microdioritic inclusions and dykes was intruded by the monzogranite of Natzwiller characterized by K-Ar ages of 344 and 351 ± 11 Ma [11] and an ^{40}Ar-^{39}Ar plateau at 338 ± 1.3 Ma on biotite [26]. Microdioritic inclusions of this granite (VN 51, 53) and a microgranitic dyke associated with it (VN 5) displays also C_p low blocking temperature components and C_n intermediate temperature (580 °C) components. Consequently, the C magnetizations were emplaced in the time range 340–325 Ma. Consistent K-Ar ages of 342 and 330 Ma were respectively obtained on hornblende and biotite from Odenwald [17] were we have also found a few C directions.

9. Discussion

In each investigated massif, the demagnetization process has established several groups of characteristic directions, among which the A, A′, B and C directions; each of these directions is present in at least two massifs. Concerning the A′ and B directions both normal and reversed, antiparallel directions were obtained over an area several hundred kilometers wide. These results lead to the following preliminary conclusions:

(1) In rocks emplaced in the range 362–340 Ma and cooled up to 325 Ma, most of the magnetizations from Odenwald, Spessart and Pfalz are Late Carboniferous–Early Triassic overprints with B, A′ and A directions often carried by secondary hematite; the northern Vosges are less remagnetized.

(2) The existence of different time dependent paleomagnetic directions with the sequence C-B-A′-A in the same massif or area has to be interpreted in terms of regional or global tectonics.

(3) The same directions found in the different massifs confirms the conclusion drawn from the magnetic anomaly maps: no significant relative rotations affected northern Vosges, Odenwald and Spessart since the Early Carboniferous.

9.1. The problem of the C directions

Apparent Virtual Geomagnetic North Poles are computed for the mean A, A′, B directions (Fig. 4). By reason of the unclear polarity of the oldest C directions both, the C and C′ poles are represented.

The low unblocking temperatures of the C_p components in diorites and schists and the negative fold test in the late Devonian–Visean schists north of the plutons indicate that they were em-

placed after the C_n, which are considered as thermomagnetic components carried by primary magnetite. The C_n directions have probably taken place during crystallisation of the granodiorites, around 340–335 Ma, and the C_p later as overprints. Two cases have to be considered:

(1) The C_p low temperature overprints are characteristic of the magnetic field in the time range 330–325 Ma, during unroofing and cooling of the ignous basement, [11,16,17].

(2) They are much later overprints, with directions intermediate to the C_n and the Jurassic to Present directions, and consequently are meaningless. This interpretation is suspected for site VN 61 in which the low-temperature C_p direction seems to lay on a great circle intermediate to the higher temperature (580°) C_n direction and the present field direction.

Due to this undetermination and to the unclear polarity of the C directions, three possible C-B apparent polar wander paths are proposed:

— The path 1 (C_n–C_p–B) takes into account both the C_n and C_p poles (solution 1). The C_n directions are the oldest, the C_p are later overprints. The deviation from C_n to C_p is due either to a latitudinal drift from about 15° N to 15° S corresponding to at least 3300 km, without any change in direction, or to southward tilting of the Champ du Feu massif and the Spessart, the C_n directions being tilted C_p directions. Then the deviation from C_p to B has to be interpreted as resulting from a 80° clockwise rotation without change in paleolatitude of the Saxothuringian and Moldanubian blocks.

— The path 2 (C_n'–C_p–B) takes into account the opposite pole C_n'; it implies a first very large counterclockwise rotation by about 180° (C_n'–C_p), followed by the same 80° clockwise rotation as above.

— In path 3 (C_n'–B) the C_p are considered as meaningless (solution 2). The C_n'–B deviation leads to a 85° counterclockwise rotation of the concerned massifs and the C_3'–C_n' path to an eventual previous counterclockwise rotation or southward tilting of the northern Vosges batholite.

Both, the C_n'–C_p–B–A′–A and C_n'–B–A′–A paths are characterized by a kink around the C_p or the B poles. The close inclinations of the C_n', C_p, B and C_3 directions imply no significant change in paleolatitude during the rotations. The position of the A′ poles located on the C_n'–B path questions then the age of the corresponding magnetizations; part of the A′ directions could be older than the B directions.

In all cases, the latest C_p directions are consistent with the mean C_p direction ($D = 314°$, $I = 26°$; VGP: 39° N, 251° E) measured in Late Devonian–Early Visean plutonics from the southern Moldanubian Vosges, while in Late Visean volcanics from the same area consistent C_n and C_p directions also coexist.

10. Geodynamic evolution of the Saxothuringian during the Carboniferous–Permian

10.1. Late Visean–Namurian

Two paleopositions respectively based on the C_p and C_n' directions are proposed (Fig. 5). The reconstructions do not take into account eventual deformations within the Saxothuringian. It is only supposed that the NNE–SSW Rhinegraben wrench faulting was not yet settled. In both cases the area was located between 5 and 20° S.

In the latest Visean–Namurian, during the major Variscan convergence phase (Sudetian), the Saxothuringian underwent a large rotation of at least 80°, clockwise or counterclockwise. Usually block rotations are accompanied by wrench faulting and the directions of rotation and of strike slip have to be consistent [27]. In the investigated area, Phyllite zone, Odenwald, Vosges [5,28,29], the unclear absolute timing of the dextral E–W to SW–NE wrench faults overprinted by NNE–SSW left-lateral strike slip in the Late Visean–Namurian precludes to draw a definitive conclusion, but we are tempted to relate the dextral movements with a counterclockwise rotation (path 2 or 3). The Moldanubian southern Vosges and the northern Central Massif have undergone the same motions.

10.2. Namurian–Westphalian

In the northern Vosges and in the Schwarzwald took respectively place the Kagenfels rhyolite and the Munstertal ignimbrite, both having B directions [8]. B overprints affected most Early Carboniferous plutonics of the Odenwald.

The deviation in declination from B to A′ is interpreted as resulting from a 40° clockwise rotation of the whole investigated area, without significant change in paleolatitude (10° S, Fig. 5). The same movement has affected Armorican Mas-

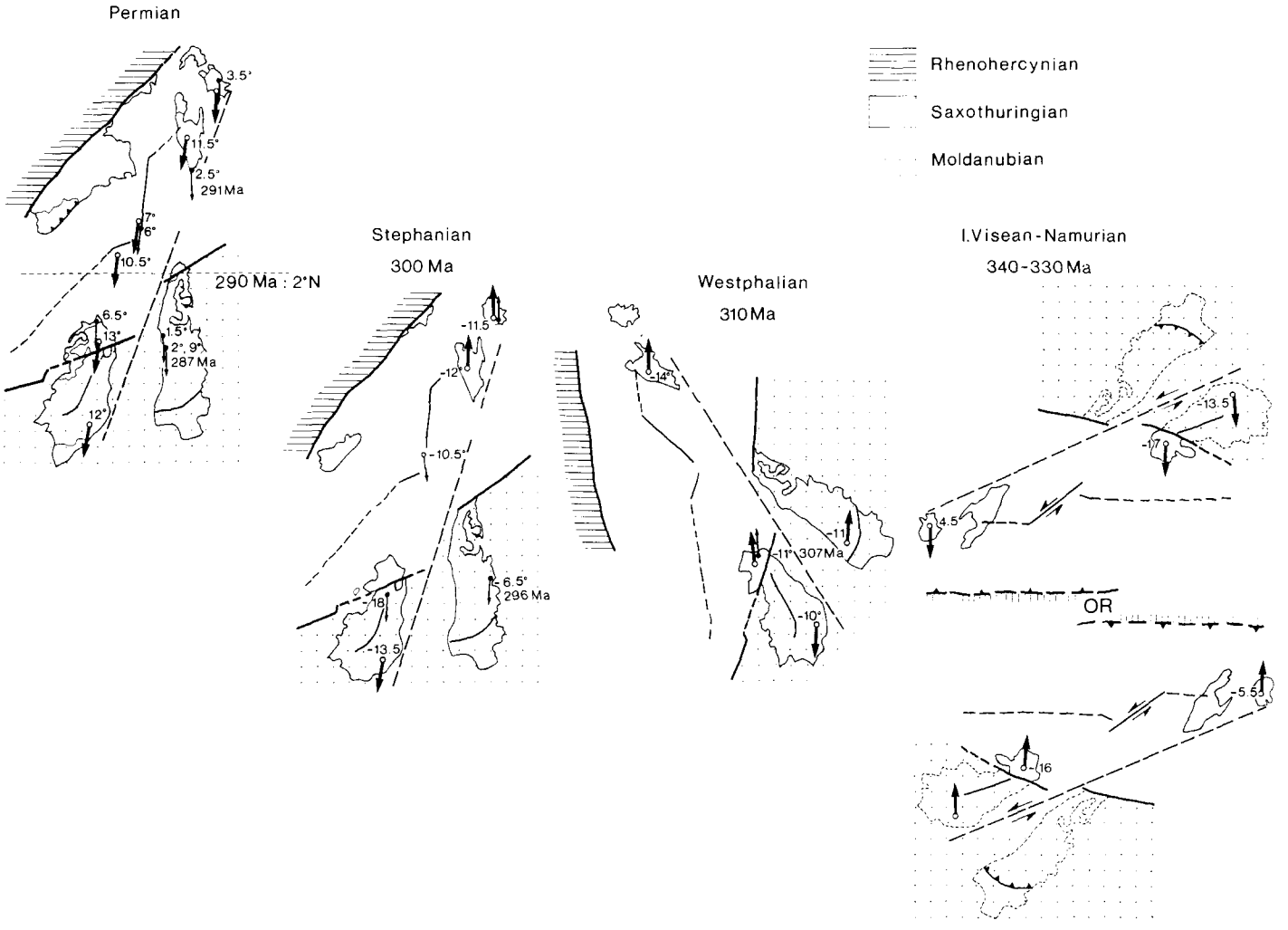


Fig. 5. Paleopositions of the upper Rhenish area during the Carboniferous–Permian derived respectively from the A, A′, B and C paleomagnetic directions obtained in the Saxothuringian (this work), and in the Moldanubian [11] (southern Vosges: in work). For the lower Visean–Namurian, two solutions, respectively based on C_p and C'_n directions are proposed. Thin arrows represent primary magnetizations obtained in one or two sites; thick arrows represent means including primary and secondary magnetizations. Numbers are paleolatitudes. The radiometric ages of the paleomagnetic sites with primary magnetizations are from Lippolt et al. [23].

sif, Ardenne, Central Massif, the Moldanubian southern Vosges, Schwarzwald [8] and the Bohemian massif (preliminary results).

According to the estimated ages of respectively 320–307 Ma of the B magnetizations, 300 Ma of the A′, the rotation has probably occurred in the range 307–300 Ma. It may be correlated with the emersion and erosion phase, leading to the Sarrebruck–Merlebach ridge, and with the change in the sedimentation at the end of the Westphalian [30]. During the same period, the Basin of Paris and the northern Central Massif have undergone a clockwise rotation of the stress field, of the same order as the paleomagnetic directions [31]. Edel [8] proposes that Northern Europe was also involved in this global Late Variscan clockwise rotation.

10.3. Stephanian–Permian

In the Stephanian, overprinting continued and dykes having A′ directions were intruded in the northern Saxothuringian. To the clockwise Westphalian–Stephanian rotation succeeded a northward latitudinal drift (path A′–A), by about 2500 km, from 10° S in the Early Stephanian to 13° N in the Late Permian–Early Trias (Figs. 4, 5). The equator was crossed at around 290 Ma. During the Permo-Triassic the whole area was affected by low-temperature hydrothermal activity producing secondary hematite associated with magnetic overprints. These Permo-Triassic overprints show a slight but systematic deviation by 5 to 10° with the Stephanian–Early Permian primary magnetizations which is in favour of a small counterclockwise rotation during the Permian. The rotation-northward drift motion continued during the Triassic as far as the Early Jurassic (Figs. 4, 5).

Acknowledgements

Thanks are due to A. Krohe for guiding in the Odenwald. R. Montigny and M. Westphal read the manuscript and made critical and fruitful comments. The work was supported by URA 323 of Centre National de Recherche Scientifique and Université Louis Pasteur of Strasbourg.

References

1 F. Volker and R. Altherr, Lower Carboniferous calc-alkaline volcanics in the northern Vosges: evidence for a destructive continental margin, Terra Cognita 7(2), 174–175, 1987.
2 T.C. Liew and A.W. Hofmann, Precambrian crustal components, plutonic associations, plate environment of the Hercynian Fold Belt of Central Europe: indications from Nd and Sr isotopic study, Contrib. Miner. Petrol. 98, 129–138, 1988.
3 F. Kossmatt, Gliederung des varistischen Gebirgsbaues, Abh. Sächs. Geol. L.-A. 1, 1–39, 1927.
4 H.D. Pflug and E. Reitz, Palynology in metamorphic rocks: indication of early land plants, Naturwissenschaften 74, 386, 1987.
5 F. Wickert and G.H. Eisbacher, Two-sided Variscan thrust tectonics in the Vosges Mountains, northeastern France, Geodin. Acta 2, 101–120, 1988.
6 J.B. Edel and M. Coulon, Late Hercynian remagnetizations of Tournaisian series from the Laval syncline, Armorican massif, France, Earth Planet. Sci. Lett. 68, 343–350, 1984.
7 J.B. Edel and M. Coulon, A paleomagnetic cross section through the Ardenne and Brabant massifs (France-Belgium), J. Geophys. 61, 21–29, 1987.
8 J.B. Edel, Paleopositions of the western Europe Hercynides during the late Carboniferous deduced from paleomagnetic data: consequences for "stable" Europe, Tectonophysics 139, 31–42, 1987.
9 J.B. Edel, Paleomagnetic evolution of the Central Massif (France) during the Carboniferous, Earth Planet. Sci. Lett. 82, 180–192, 1987.
10 J.B. Edel, M. Lacaze and M. Westphal, Paleomagnetism in the northeastern Central Massif (France), evidence for Carboniferous rotations of the orogenic belt, Earth Planet. Sci. Lett. 55, 48–52, 1981.
11 J.B. Edel, R. Montigny, J.Y. Royer, R. Thuizat and F. Trolard, Paleomagnetic investigations and K-Ar dating on the Variscan plutonic massif of the Champ du Feu and its volcanic sedimentary environment, northern Vosges, France, Tectonophysics 122, 165–185, 1986.
12 J.B. Edel, M. Coulon and M.P. Hernot, Mise en évidence par le paléomagnétisme d'une importante rotation antihoraire des Vosges méridionales entre le Viséen terminal et le Westphalien supérieur, Tectonophysics 106, 239–257, 1984.
13 J.G. Blanalt and F. Lillié, Données nouvelles sur la stratigraphie des terrains sédimentaires dévono-dinantiens de la vallée de la Bruche (Vosges septentrionales), Sci. Géol. Bull. 26, 1, 69–74, 1973.
14 E. Reitz and F. Wickert, Late Cambrian to early Ordovician acritarchs from the Villé Unit, Northern Vosges Mountains (France), N. Jahrb. Geol. Palaeontol. Monatsh. 6, 375–384, 1989.
15 F. Wickert, R. Altherr and M. Deutsch, Polyphase Variscan tectonics and metamorphism along a segment of the Saxothuringian-Moldanubian boundary: the Baden-Baden zone, northern Schwarzwald (F.R.G.), Geol. Rundsch., in press.
16 R. Montigny, C. Schneider, J.Y. Royer and R. Thuizat, K-Ar dating of some plutonics of the Vosges (France), Terra Cognita 3, 201, 1983.
17 H. Kreuzer and W. Harre, K/Ar Altersbestimmungen an

Hornblenden und Biotiten des Kristallinen Odenwaldes, Aufschluss Sonderh. 27, 71–77, 1975.
18 H. Kirsch, B. Kober and H.J. Lippolt, Age of intrusion and rapid cooling of the Frankenstein gabbro (Odenwald, SW-Germany) evidenced by $^{40}Ar/^{39}Ar$ and single zircon $^{207}Pb/^{206}Pb$ measurements, Geol. Rundsch. 77, 3, 693–711, 1988.
19 H. Kreuzer, H. Lenz, W. Harre, S. Matthes, M. Okrusch and P. Richter, Zur Alterstellung der Rotgneise im Spessart, Rb:Sr Gesamtsgesteindatierungen, Geol. Jahrb. A9, 69–88, 1973.
20 H.J. Lippolt, Nachweis altpaläozoischer Primäralter (Rb-Sr) und karbonischer Abkühlungsalter (K-Ar) der Muskovit-Biotit-Gneise des Spessarts und der Biotit-Gneise des Böllsteiner Odenwaldes, Geol. Rundsch. 75, 569–583, 1986.
21 M.P. Hernot, Etude paléomagnétique de quelques formations volcaniques et plutoniques carbonifères des Vosges, Thèse Doct., Strasbourg, 160 pp, 1983.
22 A. Roche, N. Saucier and J. Lacaze, Etude paléomagnétique des roches volcaniques permiènnes de la région du Nideck-Donon, Bull. Serv. Carte Géol. Alsace-Lorraine 15, 59–68, 1962.
23 H.J. Lippolt, H. Schleicher and I. Raczek, Rb-Sr systematics of Permian volcanites in the Schwarzwald (SW-Germany), Contrib. Mineral. Petrol. 84, 272–280, 1983.
24 H.J. Konrad and A.E.M. Nairn, The palaeomagnetism of the Permian rocks of the Black Forest, Germany, Geophys. J. R. Astron. Soc. 27, 369–382, 1972.
25 W. Bosum, H. Geipel and W. Weinelt, Magnetische Untersuchungen im Kristallinen Spessart und ihre geophysikalisch-geologische Deutung, Geol. Bavarica 67, 11–34, 1973.
26 R. Montigny and R. Thuizat, K-Ar and Ar-Ar ages on crystalline rocks of the Vosges, Terra Abstr., 1, 352, 1989.
27 J. Jackson and D. McKenzie, The relationship between plate motions and seismic moment tensors, and the rate of active deformation in the Mediterranean and Middle East, Geophys. J. 93, 45–73, 1988.
28 P. Fluck, J.B. Edel, C. Gagny, R. Montigny, A. Piqué, J.L. Schneider and H. Whitechurch, Carte synthétique et géotraverse N–S de la chaîne varisque des Vosges (France). Synthèse des travaux éffectués depuis deux décénies, C. R. Acad. Sci. Paris 309, Sér. II, 907–912, 1989.
29 J.B. Edel and P. Fluck, The upper Rhenish Shield basement (Vosges, Upper Rhinegraben and Schwarzwald): main structural features deduced from magnetic, gravimetric and geological data, Tectonophysics 169, 303–316, 1989.
30 A. Schäfer, Variscan molasse in the Saar-Nahe Basin (W-Germany), Upper Carboniferous and Lower Permian, Geol. Rundsch. 78, 499–524, 1989.
31 C. Castaing, A. Autran, G. Delpont and M. Turland, Evolution des différents contextes tectoniques et géodynamiques du socle du Bassin de Paris depuis le Protérozoïque, Doc. BRGM 81–2, 163–184, 1984.
32 V. Courtillot, P. Chambon, J.P. Brun, P. Rochette and P. Matte, A magnetotectonic study of the Hercynian Montagne Noire, Tectonics 5, 733–752, 1986.
33 N. Merabet and L. Daly, Détermination d'un pôle paléomagnétique et mise en évidence d'aimantations à polarité normale sur les formations du Permien supérieur du Massif des Maures (France), Earth Planet. Sci. Lett. 80, 156–166, 1986.
34 F. Certier, Etude paléomagnétique du Trias rhénan, Mém. DEA, 32 pp., 1988.
35 R.J.E. Johnson, R. Van Der Voo and W. Lowrie, Paleomagnetism and late diagenesis of Jurassic carbonates from the Jura Mountains, Switzerland and France, Geol. Soc. Am. Bull. 95, 478–488, 1984.
36 M. Kadzialko-Hofmokl, J. Kruczyk and M. Westphal, Paleomagnetism of Jurassic sediments from the western border of the Rhinegraben, Alsace (France), J. Geophys. 62, 102–108, 1988.